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(54) **EXPRESSION SYSTEMS FOR THE
ALPHA6BETA4 NICOTINIC
ACETYLCHOLINE RECEPTOR AND
METHODS OF USE THEREOF**

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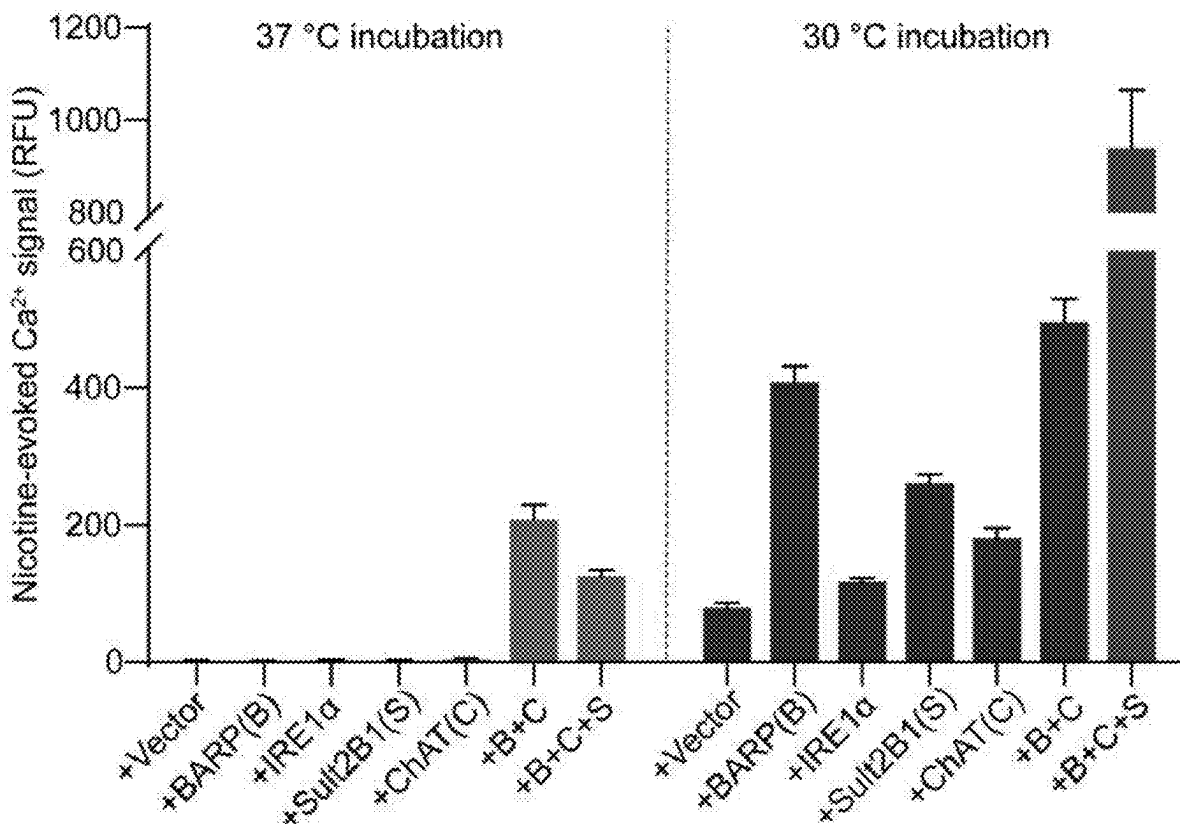
2333/91194 (2013.01)

(57)

ABSTRACT

Disclosed herein are isolated recombinant cells for the
expression of $\alpha 6 \beta 4$ nicotinic acetylcholine receptor
(nAChR) and methods of use thereof.

Specification includes a Sequence Listing.



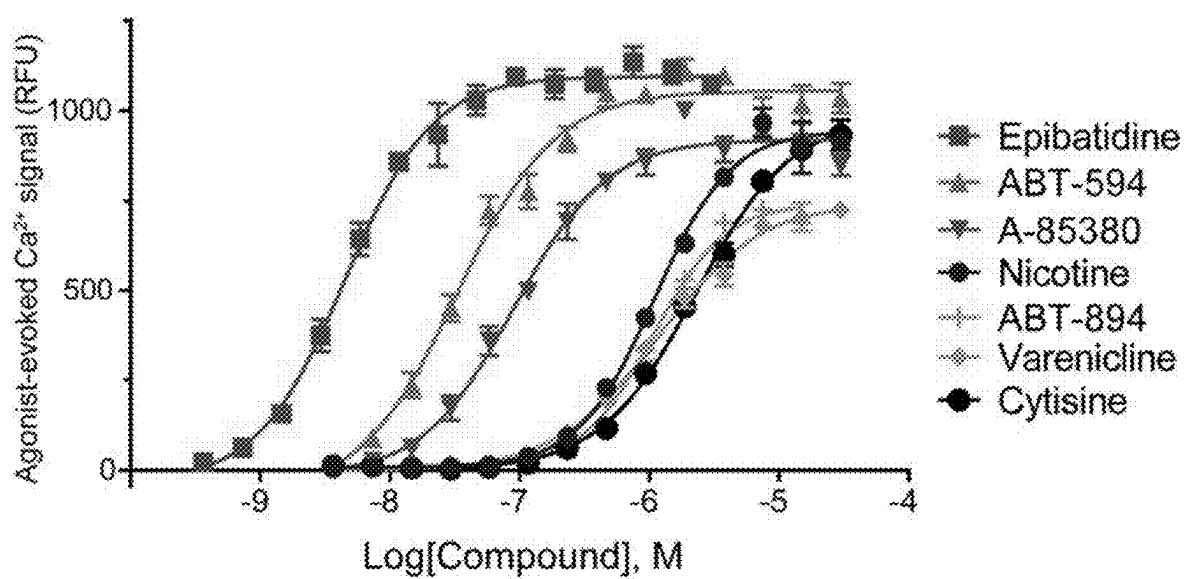


Figure 1

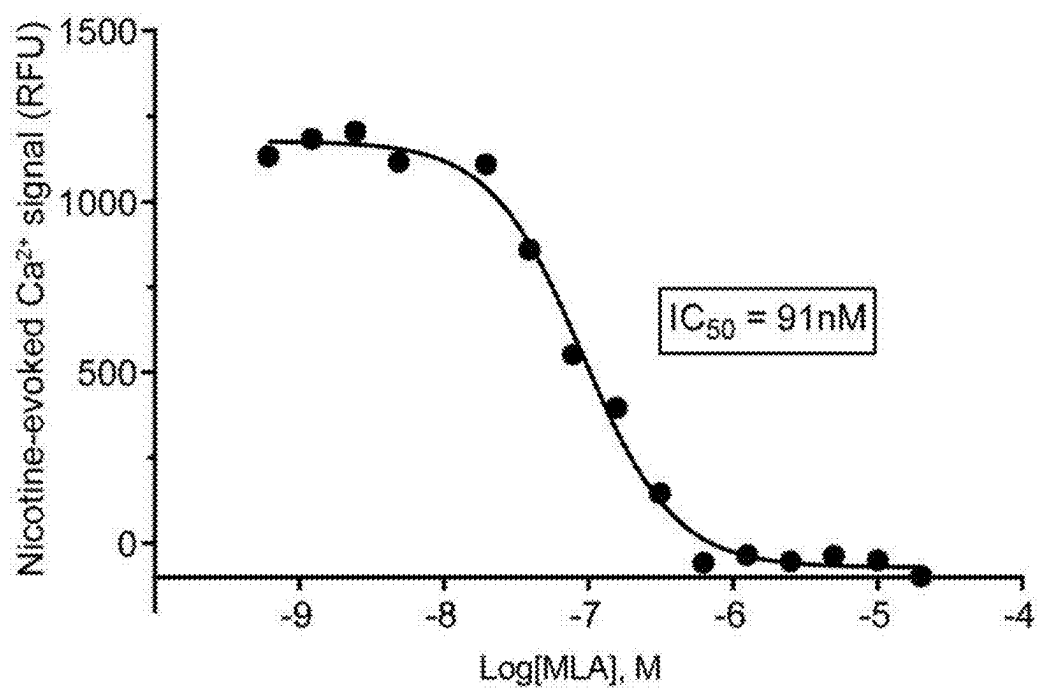


Figure 2

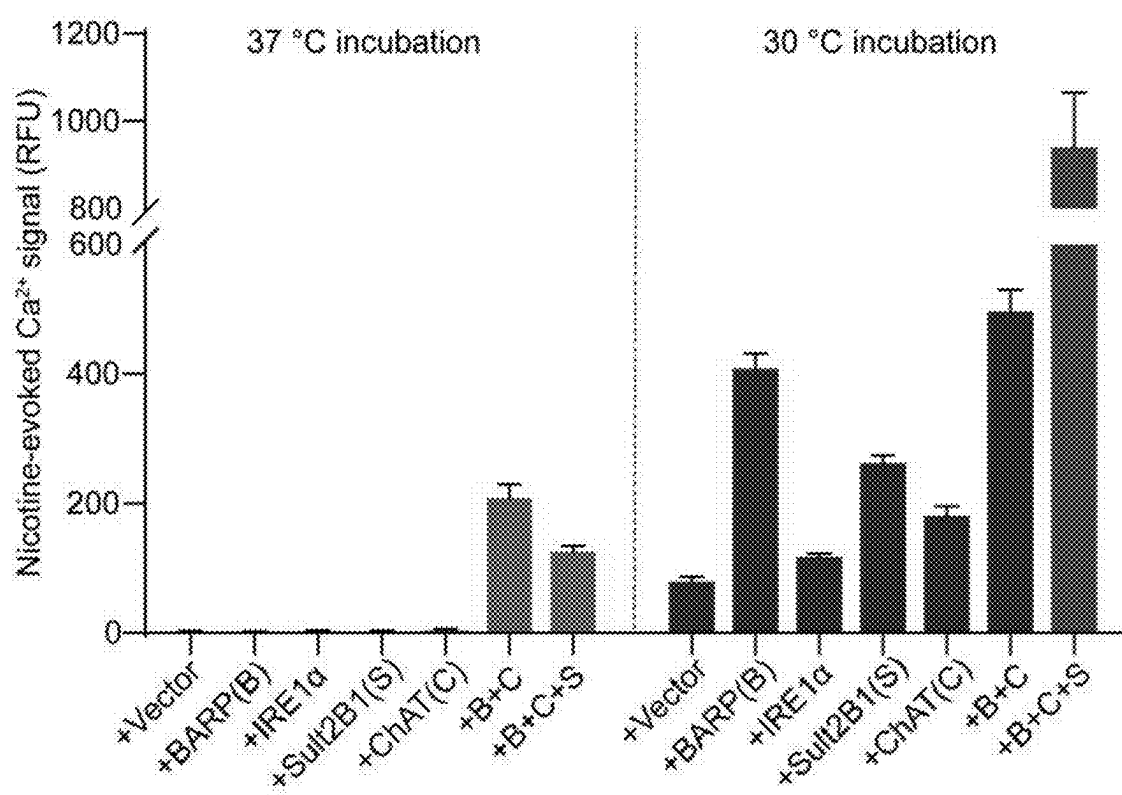


Figure 3

EXPRESSION SYSTEMS FOR THE ALPHA6BETA4 NICOTINIC ACETYLCHOLINE RECEPTOR AND METHODS OF USE THEREOF

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of U.S. Provisional Application 63/178,835, filed on Apr. 23, 2021, which is incorporated by reference herein in its entirety.

FIELD OF INVENTION

[0002] This invention relates to isolated recombinant cells for the expression of $\alpha 6\beta 4$ nicotinic acetylcholine receptor (nAChR) and methods of use thereof.

REFERENCE TO SEQUENCE LISTING SUBMITTED ELECTRONICALLY

[0003] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Feb. 28, 2022, is named PRD4127WOPCT1_SL.txt and is 24,576 bytes in size.

BACKGROUND OF INVENTION

[0004] Nicotine has analgesic properties owing to actions on nicotinic acetylcholine receptors (nAChRs) (Hone et al., *Nicotinic Acetylcholine Receptors in Neuropathic and Inflammatory Pain*, FEBS Lett. 2018 April; 592 (7):1045-1062. doi: 10.1002/1873-3468.12884). Epibatidine, is a general nAChR agonist and a powerful analgesic (Daly et al., *Alkaloids from Frog Skin: the Discovery of Epibatidine and the Potential for Developing Novel Non-Opioid Analgesics* Nat Prod Rep. 2000 April; 17(2):131-5. doi: 10.1039/a900728h). Furthermore, a pan-nAChR agonist ABT-594 showed robust efficacy for diabetic neuropathy in a phase 2 clinical trial. Unfortunately, ABT-594 had unacceptable side effects and its precise molecular target was unknown (Rowbotham et al., *A randomized, double-blind, placebo-controlled trial evaluating the efficacy and safety of ABT-594 in patients with diabetic peripheral neuropathic pain*, Pain. 2009 December; 146(3):245-252. doi: 10.1016/j.pain.2009.06.013. Epub 2009 Jul. 24). Subsequently, a genomics screen of mRNA expression in dorsal root ganglion tissue from outbred mouse strains identified that 6 nAChR subunit levels inversely correlate with pain responses in a spare nerve injury model (Wieskopf et al., *The nicotinic $\alpha 6$ subunit gene determines variability in chronic pain sensitivity via cross-inhibition of P2X2/3 receptors*, Sci Transl Med. 2015 May 13; 7(287):287ra72. doi: 10.1126/scitranslmed.3009986). Accordingly, analgesic effects of nicotine following both inflammatory and neuropathic injuries are absent in $\alpha 6$ knockout mice. Furthermore, human postoperative pain and temporomandibular disorder are affected by a polymorphism in the CHRNA6 ($\alpha 6$) promoter (Wieskopf et al., *The nicotinic $\alpha 6$ subunit gene determines variability in chronic pain sensitivity via cross-inhibition of P2X2/3 receptors*, Sci Transl Med. 2015 May 13; 7(287):287ra72. doi: 10.1126/scitranslmed.3009986). Taken together, these results provide compelling evidence that agonism on $\alpha 6$ in spinal neuron, which pair with $\beta 4$ subunits can relieve neuropathic pain and is an appealing non-opioid target (Hone et al., *Nicotinic Acetylcholine Receptors in Neuro-*

pathic and Inflammatory Pain, FEBS Lett. 2018 April; 592(7):1045-1062. doi: 10.1002/1873-3468.12884).

[0005] Drug discovery on human $\alpha 6\beta 4$ receptors has however not been possible, as these receptors have been refractory to expression in recombinant cell lines (Yang et al., *Mysterious $\alpha 6$ -containing nAChRs: function, pharmacology, and pathophysiology*, Acta Pharmacologica Sinica, 2009 June; 30(6): 740-751)

[0006] There still is a need to develop a cell line that expresses these receptors robustly and can be used for drug discovery.

BRIEF SUMMARY OF THE INVENTION

[0007] Provided herein are isolated recombinant cells comprising: a) a heterologous nucleic acid encoding an $\alpha 6$ subunit of an $\alpha 6\beta 4$ nicotinic acetylcholine receptor (nAChR); b) a heterologous nucleic acid encoding a $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR; and c) at least one heterologous nucleic acid selected from the group consisting of a heterologous nucleic acid encoding beta associated regulatory protein (BARP), a heterologous nucleic acid encoding choline O-acetyltransferase (CHAT), and a heterologous nucleic acid encoding sulfotransferase family 2B member 1 (SULT2B1).

[0008] In one embodiment of the isolated recombinant cells, the recombinant cell comprises the heterologous nucleic acid encoding BARP, the heterologous nucleic acid encoding CHAT, and the heterologous nucleic acid encoding SULT2B1.

[0009] In a further embodiment of the isolated recombinant cells, the recombinant cell is a mammalian cell.

[0010] In a yet further embodiment of the isolated recombinant cells, the mammalian cell is selected from the group consisting of a human embryonic kidney 293T (HEK293T) cell, a HEK293F cell, a HeLa cell, a Chinese hamster ovary (CHO) cell, a NIH 3T3 cell, a MCF-7 cell, a Hep G2 cell, a baby hamster kidney (BHK) cell, and a Cos7 cell.

[0011] In a yet further embodiment of the isolated recombinant cells, the $\alpha 6$ subunit of the $\alpha 6\beta 4$ nAChR comprises an amino acid sequence with at least 95% identity to the amino acid sequence of SEQ ID NO: 1.

[0012] In a yet further embodiment of the isolated recombinant cells, the $\beta 4$ subunit of the $\alpha 6\beta 4$ nAChR comprises an amino acid sequence with at least 95% identity to the amino acid sequence of SEQ ID NO: 2.

[0013] In a yet further embodiment of the isolated recombinant cells, the BARP comprises an amino acid sequence with at least 95% identity to the amino acid sequence of SEQ ID NO: 3.

[0014] In a yet further embodiment of the isolated recombinant cells, the CHAT comprises an amino acid sequence with at least 95% identity to the amino acid sequence of SEQ ID NO: 4.

[0015] In a yet further embodiment of the isolated recombinant cells, the SULT2B1 comprises an amino acid sequence with at least 95% identity to the amino acid sequence of SEQ ID NO: 5.

[0016] Further provided herein are methods for identifying agonists, antagonists, or positive allosteric modulators of $\alpha 6\beta 4$ nAChR, the method comprising: a) contacting the isolated recombinant cell described above with an agent; and b) determining the activity of the $\alpha 6\beta 4$ nAChR of the isolated recombinant cell, wherein the agent is identified as an agonist or positive allosteric modulator (PAM) if the

agent enhances the activity of the $\alpha 6 \beta 4$ nAChR and the agent is identified as an antagonist if the agent decreases the activity of the $\alpha 6 \beta 4$ nAChR as compared to the activity of the $\alpha 6 \beta 4$ nAChR when the isolated recombinant cell was not contacted with the agent.

[0017] In a further embodiment of the method, step b) comprises determining calcium flux of the isolated recombinant cell, wherein the agent is identified as an agonist if the agent enhances the calcium flux as compared to the calcium flux when the isolated recombinant cell was not contacted with the agent.

[0018] In a yet further embodiment of the method, step b) comprises determining calcium flux and nicotine-evoked calcium flux of the isolated recombinant cell, wherein the agent is identified as a PAM if the agent does not enhance calcium flux and enhances the nicotine-evoked calcium flux as compared to the calcium flux and nicotine-evoked calcium flux when the isolated recombinant cell was not contacted with the agent.

[0019] In a yet further embodiment of the method, step b) comprises determining nicotine-evoked calcium flux of the isolated recombinant cell, wherein the agent is identified as an antagonist if the agent decreases the nicotine-evoked calcium flux as compared to the nicotine-evoked calcium flux when the isolated recombinant cell was not contacted with the agent.

[0020] In a yet further embodiment of the method, the isolated recombinant cell is incubated at about 25° C.-35° C. for about 20-50 hours prior to being contacted with the agent.

[0021] In a yet further embodiment of the method, the agent is a small molecule or peptide.

[0022] Yet further provided herein is a kit comprising (i) the isolated recombinant cell as described above and (ii) instructions for use.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. 1 is a plot of FLIPR signal of the recombinant cells after incubation with agonists of $\alpha 6 \beta 4$ nAChR at various concentrations.

[0024] FIG. 2 is a plot of FLIPR signals of the recombinant cells after first incubation with methyllycaconitine (MLA) (a known antagonist of $\alpha 6 \beta 4$ nAChR) at various concentrations and second incubation with nicotine.

[0025] FIG. 3 is a graph showing functional expression of $\alpha 6 \beta 4$ nAChR when co-expressed with different chaperon proteins.

DETAILED DESCRIPTION OF THE INVENTION

[0026] Various publications, articles and patents are cited or described in the background and throughout the specification; each of these references is herein incorporated by reference in its entirety. Discussion of documents, acts, materials, devices, articles or the like which has been included in the present specification is for the purpose of providing context for the invention. Such discussion is not an admission that any or all of these matters form part of the prior art with respect to any inventions disclosed or claimed.

[0027] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this

invention pertains. Otherwise, certain terms used herein have the meanings as set forth in the specification.

[0028] It must be noted that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural reference unless the context clearly dictates otherwise.

[0029] Unless otherwise stated, any numerical values, such as a concentration or a concentration range described herein, are to be understood as being modified in all instances by the term “about.” Thus, a numerical value typically includes $\pm 10\%$ of the recited value. For example, a concentration of 1 mg/mL includes 0.9 mg/mL to 1.1 mg/mL. Likewise, a concentration range of 1% to 10% (w/v) includes 0.9% (w/v) to 11% (w/v). As used herein, the use of a numerical range expressly includes all possible sub-ranges, all individual numerical values within that range, including integers within such ranges and fractions of the values unless the context clearly indicates otherwise.

[0030] Unless otherwise indicated, the term “at least” preceding a series of elements is to be understood to refer to every element in the series. Those skilled in the art will recognize or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the invention described herein. Such equivalents are intended to be encompassed by the invention.

[0031] As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” “contains” or “containing,” or any other variation thereof, will be understood to imply the inclusion of a stated integer or group of integers but not the exclusion of any other integer or group of integers and are intended to be non-exclusive or open-ended. For example, a composition, a mixture, a process, a method, an article, or an apparatus that comprises a list of elements is not necessarily limited to only those elements but can include other elements not expressly listed or inherent to such composition, mixture, process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0032] As used herein, the conjunctive term “and/or” between multiple recited elements is understood as encompassing both individual and combined options. For instance, where two elements are conjoined by “and/or,” a first option refers to the applicability of the first element without the second. A second option refers to the applicability of the second element without the first. A third option refers to the applicability of the first and second elements together. Any one of these options is understood to fall within the meaning, and therefore satisfy the requirement of the term “and/or” as used herein. Concurrent applicability of more than one of the options is also understood to fall within the meaning, and therefore satisfy the requirement of the term “and/or.”

[0033] As used herein, the term “consists of,” or variations such as “consist of” or “consisting of,” as used throughout the specification and claims, indicate the inclusion of any recited integer or group of integers, but that no additional integer or group of integers can be added to the specified method, structure, or composition.

[0034] As used herein, the term “consists essentially of,” or variations such as “consist essentially of” or “consisting essentially of,” as used throughout the specification and

claims, indicate the inclusion of any recited integer or group of integers, and the optional inclusion of any recited integer or group of integers that do not materially change the basic or novel properties of the specified method, structure or composition. See M.P.E.P. § 2 111.03.

[0035] It should also be understood that the terms “about,” “approximately,” “generally,” “substantially,” and like terms, used herein when referring to a dimension or characteristic of a component of the preferred invention, indicate that the described dimension/characteristic is not a strict boundary or parameter and does not exclude minor variations therefrom that are functionally the same or similar, as would be understood by one having ordinary skill in the art. At a minimum, such references that include a numerical parameter would include variations that, using mathematical and industrial principles accepted in the art (e.g., rounding, measurement or other systematic errors, manufacturing tolerances, etc.), would not vary the least significant digit.

[0036] The terms “identical” or percent “identity,” in the context of two or more nucleic acids or polypeptide sequences, refer to two or more sequences or subsequences that are the same or have a specified percentage of amino acid residues or nucleotides that are the same, when compared and aligned for maximum correspondence, as measured using one of the following sequence comparison algorithms or by visual inspection. For sequence comparison, typically one sequence acts as a reference sequence, to which test sequences are compared. When using a sequence comparison algorithm, test and reference sequences are input into a computer, subsequence coordinates are designated, if necessary, and sequence algorithm program parameters are designated. The sequence comparison algorithm then calculates the percent sequence identity for the test sequence(s) relative to the reference sequence, based on the designated program parameters.

[0037] Optimal alignment of sequences for comparison can be conducted, e.g., by the local homology algorithm of Smith & Waterman, *Adv. Appl. Math.* 2:482 (1981), by the homology alignment algorithm of Needleman & Wunsch, *J. Mol. Biol.* 48:443 (1970), by the search for similarity method of Pearson & Lipman, *Proc. Nat'l. Acad. Sci. USA* 85:2444 (1988), by computerized implementations of these algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Dr., Madison, WI), or by visual inspection (see generally, *Current Protocols in Molecular Biology*, F. M. Ausubel et al., eds., *Current Protocols*, a joint venture between Greene Publishing Associates, Inc. and John Wiley & Sons, Inc., (1995 Supplement) (Ausubel)).

[0038] Examples of algorithms that are suitable for determining percent sequence identity and sequence similarity are the BLAST and BLAST 2.0 algorithms, which are described in Altschul et al. (1990) *J. Mol. Biol.* 215:403-410 and Altschul et al. (1997) *Nucleic Acids Res.* 25:3389-3402, respectively. Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information. This algorithm involves first identifying high scoring sequence pairs (HSPs) by identifying short words of length W in the query sequence, which either match or satisfy some positive-valued threshold score T when aligned with a word of the same length in a database sequence. T is referred to as the neighborhood word score threshold (Altschul et al, supra). These initial neighborhood word hits act as seeds for initiating searches to find longer

HSPs containing them. The word hits are then extended in both directions along each sequence for as far as the cumulative alignment score can be increased.

[0039] Cumulative scores are calculated using, for nucleotide sequences, the parameters M (reward score for a pair of matching residues; always >0) and N (penalty score for mismatching residues; always <0). For amino acid sequences, a scoring matrix is used to calculate the cumulative score. Extension of the word hits in each direction are halted when: the cumulative alignment score falls off by the quantity X from its maximum achieved value; the cumulative score goes to zero or below, due to the accumulation of one or more negative-scoring residue alignments; or the end of either sequence is reached. The BLAST algorithm parameters W, T, and X determine the sensitivity and speed of the alignment. The BLASTN program (for nucleotide sequences) uses as defaults a wordlength (W) of 11, an expectation (E) of 10, M=5, N=-4, and a comparison of both strands. For amino acid sequences, the BLASTP program uses as defaults a wordlength (W) of 3, an expectation (E) of 10, and the BLOSUM62 scoring matrix (see Henikoff & Henikoff, *Proc. Natl. Acad. Sci. USA* 89:10915 (1992)).

[0040] In addition to calculating percent sequence identity, the BLAST algorithm also performs a statistical analysis of the similarity between two sequences (see, e.g., Karlin & Altschul, *Proc. Nat'l. Acad. Sci. USA* 90:5873-5787 (1993)). One measure of similarity provided by the BLAST algorithm is the smallest sum probability (P(N)), which provides an indication of the probability by which a match between two nucleotide or amino acid sequences would occur by chance. For example, a nucleic acid is considered similar to a reference sequence if the smallest sum probability in a comparison of the test nucleic acid to the reference nucleic acid is less than about 0.1, more preferably less than about 0.01, and most preferably less than about 0.001.

[0041] A further indication that two nucleic acid sequences or polypeptides are substantially identical is that the polypeptide encoded by the first nucleic acid is immunologically cross reactive with the polypeptide encoded by the second nucleic acid, as described below. Thus, a polypeptide is typically substantially identical to a second polypeptide, for example, where the two peptides differ only by conservative substitutions. Another indication that two nucleic acid sequences are substantially identical is that the two molecules hybridize to each other under stringent conditions.

[0042] As used herein, the term “polynucleotide,” synonymously referred to as “nucleic acid molecule,” “nucleotides” or “nucleic acids,” refers to any polyribonucleotide or polydeoxyribonucleotide, which can be unmodified RNA or DNA or modified RNA or DNA. “Polynucleotides” include, without limitation single- and double-stranded DNA, DNA that is a mixture of single- and double-stranded regions, single- and double-stranded RNA, and RNA that is mixture of single- and double-stranded regions, hybrid molecules comprising DNA and RNA that can be single-stranded or, more typically, double-stranded or a mixture of single- and double-stranded regions. In addition, “polynucleotide” refers to triple-stranded regions comprising RNA or DNA or both RNA and DNA. The term polynucleotide also includes DNAs or RNAs containing one or more modified bases and DNAs or RNAs with backbones modified for stability or for other reasons. “Modified” bases include, for example, trity-

lated bases and unusual bases such as inosine. A variety of modifications can be made to DNA and RNA; thus, “polynucleotide” embraces chemically, enzymatically or metabolically modified forms of polynucleotides as typically found in nature, as well as the chemical forms of DNA and RNA characteristic of viruses and cells. “Polynucleotide” also embraces relatively short nucleic acid chains, often referred to as oligonucleotides.

[0043] As used herein, the terms “peptide,” “polypeptide,” or “protein” can refer to a molecule comprised of amino acids and can be recognized as a protein by those of skill in the art. The conventional one-letter or three-letter code for amino acid residues is used herein. The terms “peptide,” “polypeptide,” and “protein” can be used interchangeably herein to refer to polymers of amino acids of any length. The polymer can be linear or branched, it can comprise modified amino acids, and it can be interrupted by non-amino acids. The terms also encompass an amino acid polymer that has been modified naturally or by intervention; for example, disulfide bond formation, glycosylation, lipidation, acetylation, phosphorylation, or any other manipulation or modification, such as conjugation with a labeling component. Also included within the definition are, for example, polypeptides containing one or more analogs of an amino acid (including, for example, unnatural amino acids, etc.), as well as other modifications known in the art.

[0044] The peptide sequences described herein are written according to the usual convention whereby the N-terminal region of the peptide is on the left and the C-terminal region is on the right. Although isomeric forms of the amino acids are known, it is the L-form of the amino acid that is represented unless otherwise expressly indicated.

Isolated Recombinant Cells and Methods of Making Recombinant Cells

[0045] The inventions disclosed herein are based, at least in part, on the unexpected finding that co-expressing certain chaperone proteins with $\alpha 6\beta 4$ nicotinic acetylcholine receptor (nAChR) in cells generated cells that highly express $\alpha 6\delta 4$ nicotinic acetylcholine receptor (nAChR), making the cells useful for drug discovery. Provided herein are methods of making recombinant cells expressing $\alpha 6\beta 4$ nicotinic acetylcholine receptor (nAChR) and isolated recombinant cells for the expression of $\alpha 6\beta 4$ nicotinic acetylcholine receptor (nAChR).

[0046] As used herein, the terms “ $\alpha 6\beta 4$ nicotinic acetylcholine receptor,” “ $\alpha 6\beta 4$ nAChR,” “ $\alpha 6\beta 4$ nicotinic acetylcholine receptor,” and “ $\alpha 6\beta 4$ nAChR” are used interchangeably and refer to the $\alpha 6\beta 4$ nicotinic acetylcholine receptor protein, preferably the human $\alpha 6\beta 4$ nAChR, which is a member of a protein family of cholinergic receptors.

[0047] $\alpha 6\beta 4$ nAChR is a ligand-gated ion channel composed of $\alpha 6$ and $\beta 4$ subunits. The $\alpha 6$ subunit is encoded by the gene CHRNA6 (NM_004198) and the $\beta 4$ subunit is encoded by the gene CHRNA4 (NM_000750). When expressed together, the subunits co-assemble to form a heteromeric nAChR (See e.g., Hone et al., Nicotinic acetylcholine receptors in dorsal root ganglion neurons include the $\alpha 6\beta 4^*$ subtype, *FASEB J.* 2012; 26(2):917-926).

[0048] “Recombinant cells” refers to one or more individual cells as well as to a recombinant cell line in which the cells are heterologously expressing protein(s). As used herein, “heterologous expression” of a protein in a cell refers

to modifying the cell to express the protein by introducing an exogenous nucleic acid into the cell, e.g., an exogenous nucleic acid that encodes the protein to be expressed. A “heterologous nucleic acid” refers to a nucleic acid exogenous to a cell that is introduced into the cell. In some embodiments, the heterologous nucleic acid is DNA. In some embodiments, the heterologous nucleic acid is RNA. Heterologous expression of a protein in a cell can be achieved using a variety of methods. For example, an expression vector comprising a nucleic acid encoding the protein that is operably linked to a nucleic acid encoding a promoter capable of driving expression of the protein (e.g., a constitutive promoter) may be introduced into the cell.

[0049] The term “expression” as used herein, refers to the biosynthesis of a gene product. The term encompasses the transcription of a gene into RNA. The term also encompasses translation of RNA into one or more polypeptides, and further encompasses all naturally occurring post-transcriptional and post-translational modifications.

[0050] In a general aspect, the invention relates to methods of making or generating cells expressing $\alpha 6\beta 4$ nicotinic acetylcholine receptor (nAChR) that are useful for drug discovery. In one embodiment, the method comprises introducing into a cell a nucleic acid encoding an $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR, a nucleic acid encoding a $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR, and a nucleic acid encoding at least one protein selected from: beta associated regulatory protein (BARP), choline O-acetyltransferase (CHAT), and sulfotransferase family 2B member 1 (SULT2B1). The nucleic acid(s) encoding the $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR, $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR, BARP, CHAT, and/or SULT2B1 can be in an expression vector (e.g., in a single expression vector or in separate expression vectors). In some embodiments, the nucleic acid encoding the $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR, the $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR, BARP, CHAT, and/or SULT2B1 is operably linked to a promoter capable of driving expression of the respective protein. In some other embodiments, each of the nucleic acids encoding the $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR, $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR, BARP, CHAT, and SULT2B1 is operably linked to a promoter capable of driving expression of the respective protein. In some embodiments, the promoter is a constitutive promoter.

[0051] In another aspect, the method comprises introducing into a cell a nucleic acid encoding an $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR, a nucleic acid encoding a $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR, and a nucleic acid encoding at least one protein selected from: BARP, CHAT, and SULT2B1, wherein the cell generated by the method expresses $\alpha 6\beta 4$ nAChR and/or the at least one protein at an increased level compared to the same cell without the nucleic acid encoding the $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR, the nucleic acid encoding the $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR, and the nucleic acid encoding the at least one protein introduced.

[0052] In another aspect, the invention relates to cells genetically modified to express at least one protein selected from: BARP, CHAT, and SULT2B1, wherein the genetically modified cell expresses the at least one protein at an increased level relative to the expression of the same protein in the unmodified cell under the same (or substantially the same) conditions. In one embodiment, the cells are genetically modified to express each of BARP, CHAT, and SULT2B1, wherein genetically modified cell expresses BARP, CHAT, and SULT2B1 at an increased level relative

to the expression of BARP, CHAT, and SULT2B1, respectively, in the unmodified cell under the same (or substantially the same) conditions.

[0053] In another aspect, the invention relates to cells genetically modified to express an $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR, a $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR, and at least one protein selected from: BARP, CHAT, and SULT2B1, wherein the genetically modified cell expresses the at least one protein at an increased level relative to the expression of the same protein in the unmodified cell under the same (or substantially the same) conditions. In one embodiment, the cells are genetically modified to express an $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR, a $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR, and each of BARP, CHAT, and SULT2B1, wherein genetically modified cell expresses BARP, CHAT, and SULT2B1 at an increased level relative to the expression of BARP, CHAT, and SULT2B1, respectively, in the unmodified cell under the same (or substantially the same) conditions.

[0054] In another aspect, the invention relates to isolated recombinant cells comprising at least one expression vector selected from the group consisting of an expression vector comprising a nucleic acid sequence encoding BARP, an expression vector comprising a nucleic acid sequence encoding CHAT, and an expression vector comprising a nucleic acid sequence encoding SULT2B1. In one embodiment, the invention relates to isolated recombinant cells comprising an expression vector comprising a nucleic acid sequence encoding beta associated regulatory protein (BARP), an expression vector comprising a nucleic acid sequence encoding choline O-acetyltransferase (CHAT), and an expression vector comprising a nucleic acid sequence encoding sulfotransferase family 2B member 1 (SULT2B1).

[0055] In another aspect, the invention relates to isolated recombinant cells comprising an expression vector comprising a nucleic acid encoding an $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR, an expression vector comprising a nucleic acid encoding a $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR, and at least one expression vector selected from the group consisting of an expression vector comprising a nucleic acid encoding BARP, an expression vector comprising a nucleic acid encoding CHAT, and an expression vector comprising a nucleic acid encoding SULT2B1. In one embodiment, the invention relates to isolated recombinant cells comprising an expression vector comprising a nucleic acid encoding an $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR, an expression vector comprising a nucleic acid encoding a $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR, an expression vector comprising a nucleic acid encoding BARP, an expression vector comprising a nucleic acid encoding CHAT, and an expression vector comprising a nucleic acid encoding SULT2B1.

[0056] In any one of the embodiments described herein, the nucleic acid(s) encoding any one or more of the $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR, $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR, BARP, CHAT, and SULT2B1 can be in a single expression vector or in separate expression vectors.

[0057] In another aspect, the invention relates to isolated recombinant cells comprising: a) a heterologous nucleic acid encoding an $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR; b) a heterologous nucleic acid encoding a $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR; and c) at least one heterologous nucleic acid selected from the group consisting of a heterologous nucleic acid encoding BARP, a heterologous nucleic acid encoding CHAT, and a heterologous nucleic acid encoding SULT2B1.

[0058] In one embodiment, the invention relates to isolated recombinant cells comprising: a) a heterologous nucleic acid encoding an $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR; b) a heterologous nucleic acid encoding a $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR; c) a heterologous nucleic acid encoding BARP; and d) a heterologous nucleic acid encoding CHAT.

[0059] In one embodiment, the invention relates to isolated recombinant cells comprising: a) a heterologous nucleic acid encoding an $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR; b) a heterologous nucleic acid encoding a $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR; c) a heterologous nucleic acid encoding BARP; and d) a heterologous nucleic acid encoding SULT2B1.

[0060] In one embodiment, the invention relates to isolated recombinant cells comprising: a) a heterologous nucleic acid encoding an $\alpha 6$ subunit of an $\alpha 6\beta 4$ nAChR; b) a heterologous nucleic acid encoding a $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR; c) a heterologous nucleic acid encoding BARP; d) a heterologous nucleic acid encoding CHAT; and e) a heterologous nucleic acid encoding SULT2B1.

[0061] In one embodiment, the $\alpha 6$ subunit of the $\alpha 6\beta 4$ nAChR comprises an amino acid sequence with at least 85%, or at least 90%, or at least 95%, such as 95%, 96%, 97%, 98%, 99%, or 100% identity to the amino acid sequence of SEQ ID NO: 1.

SEQ ID NO: 1

MLTSKGQGFHLHGGCLWLCVFTPFKGCVCATEERLPHKLFSHYNQFIR
PVENVSDPVTVHFEVAITQLANVDEVNQIMETNLWLRIHWNDYKLRWDPM
EYDGIETLRVPADKIWKPDIVLYNNAVGDFQVEGKTKALLKYNGMITWTP
PAIFKSSCPMDITFFPFHDQNC SLKFGSWTYDKAEIDLLIIGSKVDMNDF
WENSEWEI IDASGYKHDIKYNCCEEIYTDITYSFYIRRLPMFYITINLIIP
CLFISFLTVLVFLPSDCGKVTLCISVLLSLTVFLLVITETIPSTSLVV
PLVGEYLLFTMIFVTLISVTVFVLNIHYRTPTHTMPRWVKTVFLKLLP
QVLLMRWPLDKTRGTGSDAVPRGLARRPAKGLASHGEPRLKECFCHCK
SNELATSKRRLSHQPLQWVENSEHSPVEDVINSVQFIAENMKSHNETK
EVEDDWKYVAMVVDVFLVWFIIIVCVFTAGLFLQPLLGNTGKS

[0062] In one embodiment, the $\beta 4$ subunit of the $\alpha 6\beta 4$ nAChR comprises an amino acid sequence with at least 85%, or at least 90%, or at least 95%, such as 95%, 96%, 97%, 98%, 99%, or 100% identity to the amino acid sequence of SEQ ID NO: 2:

SEQ ID NO: 2

MRRAPSLVLFFLVALCGRGNCRVANAEEKLMDLLNKTRYNNLIRPATSS
SQLISIKLQLSLAQLISVNEREQIMTTNVWLKQEWTDYRLTWNSSRYEGV
NILRIPAKRIWLPDIVLYNNADGTYEVSVYTNLIVRSNGSVLWLPPAIYK
SACKIEVKYFPFDQONCTLKFRSWTYDHTIDMVLMTPTASMDDFTPSGE
WDIVALPGRRTVNPQDPSYVDVTYDFIIRKPLFTYINLIIPCVLTTLLA
ILVFYLPSCDGKMTLCISVLLALTFFLLLSKIVPPTSLDVLPIGKYLM
FTMVLVTFISIVTSCVLNVHHRSPSTHTMAPWVKRCFLHKLPTFLFMKRP
GPDSSPARAFPPSKSVTKPEATATSTSPSNFYGNMYFVNPAASAKSP

-continued

AGSTPVAIPRDFWLRSRGRFRQDVQEALEGVSFIAQHMKNDDEDQSVVED
WKYVAMVVDRLFLWVFMFVCVLGTVGLFLPPLFQTHAASEGPYAAQRD

[0063] In one embodiment, the BARP comprises an amino acid sequence with at least 85%, or at least 90%, or at least 95%, such as 95%, 96%, 97%, 98%, 99%, or 100% identity to the amino acid sequence of SEQ ID NO: 3. Or, the BARP comprises an amino acid sequence with at least 60%, at least 70%, at least 80%, at least 85%, or at least 90%, or at least 95%, such as 95%, 96%, 97%, 98%, 99%, or 100% identity to the amino acid sequence of SEQ ID NO: 3 and has protein chaperone property, which includes the property of enhancing the expression of the $\alpha 6$ and $\beta 4$ subunits of $\alpha 6\beta 4$ nAChR.

SEQ ID NO: 3

MQPTATMATAATTTTTTATVALTTSWDNATGRPTAEPDILDNYVLLVV
VMSLFVGGTLVVLGVLCKRCWDVHQRLNRAEAEKTTTTYLDNGTH
PAQDPDFRGEDPECQDAETERFLSTSTGRRVSFNEAALFEQSRKTQDKG
RRYTLTEGDFHHLKNARLTHLHLPPLKIVTIECDSGEASSATTPHPATS
PKATLAIQPPGKALTGRSVGPSSALPGDPYNSAAGATDFAEISPSASSD
SGEGTSLDAGTRSTKAGGPGAAAGPEAGPGSGAGTVLQFLTRLRRHASL
DGASPYFKVKWKLEPSQRAASLDTRGSPKRHHFQRQRAASESTEQEEGD
APQEDFIQYIARAGDAVAFPHRPFLASPPALGRLEAAEAAGGASPDSP
PERGAGSAGPEQQPPLEPDAERDAGPEQAQTSYRDLWSLRASLEHAAA
SDHSSSGNDRDSVRSGDSSGSGGAAPAFPPSPAPRPKDGAEARRLLQ
MDSGYASIEGRGAGDDEPPAAPARPRSPRAWPRRPRRDYSIDEKTDALF
HEFLRHDPHFDDTPAAARHRRARHPARKQWQRGRQSDPGARAAPALAG
TPAPPAGAARPARAPLRRGDSVDGPPDGRTLGGAGDDPAIPVIEEPPGGG
GCPGSGLCVLPSGSLDKLAAGLDERLFPPLRAEPVVATPALVAAAPTSP
DHSPA

[0064] In one embodiment, the CHAT comprises an amino acid sequence with at least 85%, or at least 90%, or at least 95%, such as 95%, 96%, 97%, 98%, 99%, or 100% identity to the amino acid sequence of SEQ ID NO: 4. Or, the CHAT comprises an amino acid sequence with at least 60%, at least 70%, at least 80%, at least 85%, or at least 90%, or at least 95%, such as 95%, 96%, 97%, 98%, 99%, or 100% identity to the amino acid sequence of SEQ ID NO: 4 and has protein chaperone property, which includes the property of enhancing the expression of the $\alpha 6$ and $\beta 4$ subunits of $\alpha 6\beta 4$ nAChR.

SEQ ID NO: 4

MAAKTPSSEESGLPKLPVPLQQTALATYLQCMRHLVSEEQFRKSQAIVQQ
FGAPGGLGETLQQLLERQKENTANWVSEYWLNDMYLNNRLALPVNSSPAV
IFARQHFPGETDDQLRFAASLISGLVSYKALLDSHSIPTDCAKGQLSGQPL
CMKQYYGLFSSYRLPGHTQDTLVAQNSSIMPEPEHVIVACCNQFFVLDVV
INFRRLESGDLFTQLRKIVKMASNEDERLPPIGLLTSDGRSEWAEARTVL

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VKDSTNRDSLDMIERCICLVCLDAPGGVELSDTHRALQLLHGGGYSKNGA
NRWYDKSLQFVVGDRDGTGCVVCEHSPFDGIVLVQCTEHLKHMTOSSRKL
IRADSVSELPAAPRRLRWKCSPEIQGHLASSAEKLQRIVKNLDFIVYKFDN
YGKTFIKKQKSPDAFIQVALQLAFYRLHRRLVPTYESASIRRFQEGRVD
NIRSATPEALAFVRAVTDHKAAPASEKLLLLKDAIRAQTAYTVMATGM
AIDNHLALRELARAMCKELPEMFMDETYLSNRFLVSTSQVPTTTEMFC
CYGPVVPNGYGACYNPQPETILFCISSFHSCKETSSSKFAKAVEESLIDM
RDLCSLLPPTESKPLATKEKATRPSQGHQP

[0065] In one embodiment, the SULT2B1 comprises an amino acid sequence with at least 85%, or at least 90%, or at least 95%, such as 95%, 96%, 97%, 98%, 99%, or 100% identity to the amino acid sequence of SEQ ID NO: 5. Or, the SULT2B1 comprises an amino acid sequence with at least 60%, at least 70%, at least 80%, at least 85%, or at least 90%, or at least 95%, such as 95%, 96%, 97%, 98%, 99%, or 100% identity to the amino acid sequence of SEQ ID NO: 5 and has protein chaperone property, which includes the property of enhancing the expression of the $\alpha 6$ and $\beta 4$ subunits of $\alpha 6\beta 4$ nAChR.

SEQ ID NO: 5

MDGPAEPQIPGLWDITYEDDISEISQKLPGEYFRYKGVFPFVGLYSLEIS
LAENTQDVRDDDIIFIITYPKSGTTWMIEIICLILKEGDPWSIRSVPIWER
APWCETIVGAFSLPDQYSPRLMSSHLPIQIFTKAFFSSKAKVIYMGNRPR
DVVVSLEYHYSKIAGQLKDPGTPDQFLRDLKGEVQFGSWFDHIKGWLRMK
GKDNFLFIITYEELQQLQGSVERICGLGRPLGKEALGSVVAHSTFSAMK
ANTMSNYTLPPSLLDHRRGAFLRKGVCGDWKNHFTVAQSEAFDRAYRKQ
MRGMPTFPWDEDPEEDGSPDEPSPEPEPKPSLEPNTSLEREPRNPSSPS
PSPGQASETPHPRPS

[0066] Any suitable means for introducing heterologous nucleic acid into a cell can be used herein to prepare the recombinant cells disclosed herein, such as DNA transfection (e.g., via a DNA vector) and RNA transduction. In one embodiment, the heterologous nucleic acid to be introduced into cells to generate the recombinant cells are prepared by using a vector, preferably an expression vector. The term “vector” refers to a nucleic acid molecule capable of transporting another nucleic acid to which it has been linked. One type of vector is a “plasmid”, which refers to a circular double-stranded DNA loop into which additional DNA segments can be inserted. Another type of vector is a viral vector wherein additional DNA segments can be inserted. Expression vectors are those vectors capable of directing the expression of genes to which they are operably linked. The expression vectors used herein comprise a nucleic acid encoding a protein sequence in a form suitable for expression of the nucleic acid in a host cell. Thus, the expression vectors can include one or more regulatory sequences, such as a promoter, selected on the basis of the host cells to be used for expression, operably linked to the nucleic acid sequence to be expressed. When used in reference to an expression vector, “operably linked” is intended to mean

that the nucleotide sequence of interest is linked to the regulatory sequence(s) in a manner allowing for expression of the nucleotide sequence (e.g., in an in vitro transcription/translation system or in a host cell when the vector is introduced into the host cell). It will be appreciated by those of ordinary skill in the art that the design of the expression vector can depend on such factors as the choice of the host cell to be transformed and the level of expression of protein desired as well as the intended use of the vector.

[0067] Any vector known to those skilled in the art in view of the present disclosure can be used, such as a plasmid, a cosmid, a phage vector, or a viral vector. In one embodiment, the vector is an expression vector such as a plasmid. The vector can include any element to establish a conventional function of an expression vector, for example, a promoter, ribosome binding element, terminator, enhancer, selection marker, and origin of replication. The promoter can be a constitutive, inducible or repressible promoter. A number of expression vectors capable of delivering nucleic acids to a cell are known in the art and can be used herein. Conventional cloning techniques or artificial gene synthesis can be used to generate an expression vector according to embodiments of the invention.

[0068] Any cell known to those skilled in the art in view of the present disclosure can be used for recombinant expression of $\alpha 6\beta 4$ nAChR, BARP, CHAT, and SULT2B1. In one embodiment, the recombinant cell is a mammalian cell. Suitable the mammalian cells may be selected from human embryonic kidney 293T (HEK293T) cell, HEK293F cells, Hela cells, Chinese hamster ovary (CHO) cells, NIH 3T3 cells, MCF-7 cells, Hep G2 cells, baby hamster kidney (BHK) cells, and Cos7 cells.

Methods of Identifying Agonists, Antagonists, or Positive Allosteric Modulators of $\alpha 6\beta 4$ nAChR

[0069] Further provided herein are methods of identifying agonists, antagonists, or positive allosteric modulators (PAMs) of an $\alpha 6\beta 4$ nAChR. PAMs are compounds that bind at sites on the protein surface other than the active sites, and therefore change the conformation of the protein binding sites.

[0070] In one embodiment, the method comprises culturing the isolated recombinant cells disclosed herein under conditions where the recombinant cells grow, contacting the recombinant cells with an agent, and determining if the agent is an agonist, antagonist, or PAM of the $\alpha 6\beta 4$ nAChR, wherein an agonist or PAM enhances the activity of the $\alpha 6\beta 4$ nAChR and an antagonist decreases the activity of the $\alpha 6\beta 4$ nAChR as compared to the activity of the $\alpha 6\beta 4$ nAChR in a recombinant cell that was not contacted with an agent. Agonists, as used herein, refer to molecules/compounds/peptides that serve to enhance the function of the $\alpha 6\beta 4$ nAChR. PAMs, as used herein, refer to molecules/compounds/peptides that enhance the effect of $\alpha 6\beta 4$ nAChR's response to a ligand without directly activating the receptor. As used herein, the term "enhance", "enhanced", "increase", or "increased", when used with respect to $\alpha 6\beta 4$ nAChR activity refers to an increase in the signaling through the receptor, relative to the corresponding signaling observed in a cell in which an agonist or PAM is not administered. Antagonists, as used herein, refer to molecules/compounds/peptides that serve to block, decrease, or dampen the function of the $\alpha 6\beta 4$ nAChR. In particular embodiments, the agent is a small molecule or peptide.

[0071] In one embodiment, FLIPR assay is used to identify agonists of $\alpha 6\beta 4$ nAChR mediated calcium flux. In this assay, the recombinant cells are incubated with a calcium sensitive dye (such as Ca5), exposed to a test compound, and calcium flux is imaged by FLIPR^{TETRA}. As shown in FIG. 1, agonists of $\alpha 6\beta 4$ nAChR enhances calcium flux.

[0072] In one embodiment, FLIPR assay is used to identify antagonists of nicotine-evoked $\alpha 6\beta 4$ nAChR mediated calcium flux. In this assay, the recombinant cells are incubated with a calcium sensitive dye (such as Ca5). Using a double addition protocol, the recombinant cells are exposed to test compounds during the first duration, and to nicotine (e.g., at EC₈₀ (15 μ m)) during the second duration. Thereafter, the calcium flux is imaged by FLIPR^{TETRA}. As shown in FIG. 2, antagonists of $\alpha 6\beta 4$ nAChR reduce calcium flux.

[0073] In one embodiment, FLIPR assay is used to identify compounds that positively modulate or potentiate nicotine-evoked $\alpha 6\beta 4$ nAChR mediated calcium flux. In this assay, the recombinant cells are incubated with a calcium sensitive dye (such as Ca5). Using a double addition protocol, the recombinant cells are exposed to test compounds during the first duration, and to nicotine (e.g., at EC₈₀ (15 μ m)) during the second duration. Thereafter, the calcium flux is imaged by FLIPR^{TETRA}. If a test compound enhances nicotine-evoked $\alpha 6\beta 4$ nAChR mediated calcium flux, yet does not enhances calcium flux as an agonist (determined as above), the test compound is termed a positive allosteric modulator (PAM) or potentiator of $\alpha 6\beta 4$ nAChR.

[0074] In a preferred embodiment, the cells are incubated at about 25-35° C. for about 20-50 hr prior to the FLIPR assay or other assay for measuring $\alpha 6\beta 4$ nAChR activity. In some embodiments, the cells are incubated at about 25° C., about 26° C., about 27° C., about 28° C., about 29° C., about 30° C., about 31° C., about 32° C., about 33° C., about 34° C., or about 35° C., for about 20 hr, about 25 hr, about 30 hr, about 35 hr, about 40 hr, about 45 hr, or about 50 hr prior to the assay.

Expression System and Kits

[0075] Further provided herein are expression systems and kits comprising the isolated recombinant cells. The expression systems and kits may further include instructions for use.

EXAMPLE

[0076] It will be appreciated by those skilled in the art that changes could be made to the embodiments described above without departing from the broad inventive concept thereof. It is understood, therefore, that this invention is not limited to the particular embodiments disclosed, but it is intended to cover modifications within the spirit and scope of the present invention as defined by the present description.

Material

Name	Company	Cat.No
FreeStyle 293 Expression Media	Gibco	12338018
OptiPro SFM	Gibco	12309019
FreeStyle MAX Reagent	Invitrogen	16447100
Recovery Cell Culture Freezing Media	Gibco	12648010
DMEM-high glucose	Sigma	D5671-500 ml
Defined FBS	HyClone	SH30070.03

-continued

Name	Company	Cat.No
Sodium Pyruvate	HyClone	SH30239.01
Glutamax	HyClone	35050061
DMSO	Sigma Aldrich	41648
Ca5 dye	Molecular Devices	R8186
Nicotine	Tocris	3546
DH β e	Tocris	2349
384-well pp plates	Greiner	781280
384-well poly-D-lysine plates	Corning	354663
Custom HBSS	HyClone	SH3A4475.01
Calcium Chloride	Amresco	E506-500 ML
Magnesium Chloride	Amresco	E525-500 ML
Freezing Vial	Nalgene	5000-0020

Cell Line

[0077] FreeStyle 293F cell line (ThermoFisher Scientific Cat #R79007)

Vector Constructs

[0078] pcDNA3.1- α 6
 [0079] pcDNA3.1-84
 [0080] pcDNA3.1-BARP
 [0081] pcDNA3.1-SULT2B1
 [0082] pcDNA3.1-CHAT

Culturing Media

[0083] FreeStyle 293 Expression Medium
 [0084] Transfection Reagents
 [0085] FreeStyle MAX Reagent

Planting Media

[0086] DMEM
 [0087] 10% FBS
 [0088] 1% pen-strep

Compound Dilution/Ca5 Buffer.

[0089] HEPES-buffered saline solution (500 mL) supplemented with 1 mM Mg²⁺ and 2 mM Ca²⁺

Calcium 5 (Ca5) Dye (Molecular Devices)

[0090] Calcium dye diluted at 25 $\times$ concentration in Hanks Buffered Salt Solution. Dye diluted to 1 $\times$ in HEPES assay buffer with 2 mM CaCl₂) and 1 mM MgCl₂.

Methods and Procedures

Day 1 Cell Preparation

[0091] Prepare a cell suspension of 1 $\times$ 10⁶ cells per mL by centrifuging cells and resuspending in fresh, pre-warmed media;
 [0092] With 1900 mL of cells shaking, prepare the following transfection mix in 50 ml of OptiPro SFM Medium;
 [0093] Add 2.5 mL FreeStyle MAX reagent to 47.5 mL of OptiPro SFM to an empty 50 ml tube;

[0094] In a second 50 ml tube, add 2.5 mg total DNA (α 6, β 4, BARP, CHAT, and SULT2B1 at a ratio of 1:1:2:1:1) in OptiPro SFM to a final volume of 50 ml and vortex gently;

[0095] Transfer the diluted DNA to the FreeStyle MAX reagent and gently mix the solution immediately and incubate for 10 minutes at room temperature;

[0096] Transfer the 100 ml FreeStyle MAX/DNA transfection mix to the cells and homogenize the culture;

[0097] Incubate 1 hour at 37° C. with 8% CO₂;

[0098] Pour cells into sterile 500 mL tubes;

[0099] Spin at 1000 rpm for 7 minutes;

[0100] Remove supernatant; and

[0101] Resuspend Cells in Plating Media.

Day 2 Plate Cells

[0102] Ethanol vials to decontaminate;

[0103] Pour cells into 50 ml conical tube;

[0104] Slowly add 10 ml warmed plating media drop wise;

[0105] Pipet up and down gently a few times to dislodge pellet if it exists;

[0106] Spin at 1000 RPM for 7 min, aspirate supernatant, and resuspend in 25 ml plating;

[0107] Seed 35,000 cells/well in 35 μ L (1.0 $\times$ 10⁶ cells/mL); and

[0108] Incubate plates for 24 hours at 37° C. Then transfer the plate to 30° C. with 5% CO₂ in a humidified atmosphere for 24-48 hours.

Day 3 FLIPR Assay

[0109] Wash plate with plate washer (4 washes $\times$ 100 μ L/wash) using compound dilution buffer, leaving 25 UL in each well;

[0110] Add 25 μ L of 2 $\times$ Ca6 dye to each well;

[0111] Incubate at room temperature for 1 hour;

[0112] Wash plate with plate washer (4 washes $\times$ 100 μ L/wash) using compound dilution buffer, leaving 50 UL in each well;

[0113] Transfer plates to FLIPR for compound addition. Concentration Response Curves for Agonist-Evoked Activity of α 6 β 4 nAChR

[0114] Human α 6 and β 4 subunits of α 6 β 4 nAChR were co-transfected with BARP, SULT2B1, and CHAT in HEK293T cells and incubated at 30° C. for 24-48 hours. The transfected cells were incubated for one hour at room temperature with Ca5 dye followed by stimulating with agonists for 3.5 minutes. Agonist-evoked calcium flux was measured using a FLIPR^{TETRA} imager and final FLIPR signals were averaged and plotted (FIG. 1). As shown, each of the agonists tested (epibatidine, ABT-594, A-85380, nicotine, ABT-894, varenicline, and cytosine) enhances calcium flux in a dose-dependent manner.

Concentration Response Curve for Antagonist on Nicotine-Evoked Activity of α 6 β 4 nAChR

[0115] Human α 6 and β 4 subunits of α 6 β 4 nAChR were co-transfected with BARP, SULT2B1, and CHAT in HEK293T cells and incubated at 30° C. for 24-48 hour. The transfected cells were incubated for one hour at room temperature with Ca5 dye followed by stimulation using a two-addition protocol (1st addition: with methyllycaconitine (MLA) at various concentration for 3 min; 2nd addition: with nicotine (EC₈₀, 15 μ M) for 3.5 min). Nicotine-evoked cal-

cium flux was then measured using a FLIPR^{TETRA} imager and the final FLIPR signals were averaged and plotted (FIG. 2). As shown, MLA (a known agonist of $\alpha 6\beta 4$ nAChR) decreases the nicotine-evoked calcium flux in a dose-dependent manner.

Functional Expression of $\alpha 6\beta 4$ nAChR

[0116] Human $\alpha 6$ and $\beta 4$ subunits of $\alpha 6\beta 4$ nAChR were co-transfected with specified combinations of cDNAs in HEK293T cells and incubated at either 30° C. or 37° C. for

24-48 hours. The transfected cells were incubated for one hour at room temperature with Ca5 dye followed by stimulation with E_{max} nicotine (66 M). Nicotine-evoked Ca^{2+} signal of the transfected are graphed in FIG. 3. As shown, with incubation at 37° C., co-transfection with BARP and CHAT or with BARP, CHAT, and SULT2B1 enhances functional expression of $\alpha 6\beta 4$ nAChR. With incubation at 30° C., co-transfection with BARP, CHAT, or SULT2B1, both individually or in combination, enhances functional expression of $\alpha 6\beta 4$ nAChR.

SEQUENCE LISTING

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<160> NUMBER OF SEQ ID NOS: 5

<210> SEQ ID NO 1
<211> LENGTH: 494
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 1

Met Leu Thr Ser Lys Gly Gln Gly Phe Leu His Gly Gly Leu Cys Leu
1          5          10          15

Trp Leu Cys Val Phe Thr Pro Phe Phe Lys Gly Cys Val Gly Cys Ala
          20          25          30

Thr Glu Glu Arg Leu Phe His Lys Leu Phe Ser His Tyr Asn Gln Phe
          35          40          45

Ile Arg Pro Val Glu Asn Val Ser Asp Pro Val Thr Val His Phe Glu
          50          55          60

Val Ala Ile Thr Gln Leu Ala Asn Val Asp Glu Val Asn Gln Ile Met
          65          70          75          80

Glu Thr Asn Leu Trp Leu Arg His Ile Trp Asn Asp Tyr Lys Leu Arg
          85          90          95

Trp Asp Pro Met Glu Tyr Asp Gly Ile Glu Thr Leu Arg Val Pro Ala
          100         105         110

Asp Lys Ile Trp Lys Pro Asp Ile Val Leu Tyr Asn Asn Ala Val Gly
          115         120         125

Asp Phe Gln Val Glu Gly Lys Thr Lys Ala Leu Leu Lys Tyr Asn Gly
          130         135         140

Met Ile Thr Trp Thr Pro Pro Ala Ile Phe Lys Ser Ser Cys Pro Met
          145         150         155         160

Asp Ile Thr Phe Phe Pro Phe Asp His Gln Asn Cys Ser Leu Lys Phe
          165         170         175

Gly Ser Trp Thr Tyr Asp Lys Ala Glu Ile Asp Leu Leu Ile Ile Gly
          180         185         190

Ser Lys Val Asp Met Asn Asp Phe Trp Glu Asn Ser Glu Trp Glu Ile
          195         200         205

Ile Asp Ala Ser Gly Tyr Lys His Asp Ile Lys Tyr Asn Cys Cys Glu
          210         215         220

Glu Ile Tyr Thr Asp Ile Thr Tyr Ser Phe Tyr Ile Arg Arg Leu Pro
          225         230         235         240

Met Phe Tyr Thr Ile Asn Leu Ile Ile Pro Cys Leu Phe Ile Ser Phe
          245         250         255

Leu Thr Val Leu Val Phe Tyr Leu Pro Ser Asp Cys Gly Glu Lys Val
          260         265         270

Thr Leu Cys Ile Ser Val Leu Leu Ser Leu Thr Val Phe Leu Leu Val
          275         280         285
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Ile Thr Glu Thr Ile Pro Ser Thr Ser Leu Val Val Pro Leu Val Gly
 290                      295                      300

Glu Tyr Leu Leu Phe Thr Met Ile Phe Val Thr Leu Ser Ile Val Val
305                      310                      315                      320

Thr Val Phe Val Leu Asn Ile His Tyr Arg Thr Pro Thr Thr His Thr
                      325                      330                      335

Met Pro Arg Trp Val Lys Thr Val Phe Leu Lys Leu Leu Pro Gln Val
                      340                      345                      350

Leu Leu Met Arg Trp Pro Leu Asp Lys Thr Arg Gly Thr Gly Ser Asp
                      355                      360                      365

Ala Val Pro Arg Gly Leu Ala Arg Arg Pro Ala Lys Gly Lys Leu Ala
                      370                      375                      380

Ser His Gly Glu Pro Arg His Leu Lys Glu Cys Phe His Cys His Lys
385                      390                      395                      400

Ser Asn Glu Leu Ala Thr Ser Lys Arg Arg Leu Ser His Gln Pro Leu
                      405                      410                      415

Gln Trp Val Val Glu Asn Ser Glu His Ser Pro Glu Val Glu Asp Val
                      420                      425                      430

Ile Asn Ser Val Gln Phe Ile Ala Glu Asn Met Lys Ser His Asn Glu
                      435                      440                      445

Thr Lys Glu Val Glu Asp Asp Trp Lys Tyr Val Ala Met Val Val Asp
450                      455                      460

Arg Val Phe Leu Trp Val Phe Ile Ile Val Cys Val Phe Gly Thr Ala
465                      470                      475                      480

Gly Leu Phe Leu Gln Pro Leu Leu Gly Asn Thr Gly Lys Ser
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<210> SEQ ID NO 2

<211> LENGTH: 498

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 2

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Met Arg Arg Ala Pro Ser Leu Val Leu Phe Phe Leu Val Ala Leu Cys
 1                      5                      10                      15

Gly Arg Gly Asn Cys Arg Val Ala Asn Ala Glu Glu Lys Leu Met Asp
                      20                      25                      30

Asp Leu Leu Asn Lys Thr Arg Tyr Asn Asn Leu Ile Arg Pro Ala Thr
                      35                      40                      45

Ser Ser Ser Gln Leu Ile Ser Ile Lys Leu Gln Leu Ser Leu Ala Gln
50                      55                      60

Leu Ile Ser Val Asn Glu Arg Glu Gln Ile Met Thr Thr Asn Val Trp
65                      70                      75                      80

Leu Lys Gln Glu Trp Thr Asp Tyr Arg Leu Thr Trp Asn Ser Ser Arg
                      85                      90                      95

Tyr Glu Gly Val Asn Ile Leu Arg Ile Pro Ala Lys Arg Ile Trp Leu
                      100                      105                      110

Pro Asp Ile Val Leu Tyr Asn Asn Ala Asp Gly Thr Tyr Glu Val Ser
                      115                      120                      125

Val Tyr Thr Asn Leu Ile Val Arg Ser Asn Gly Ser Val Leu Trp Leu
                      130                      135                      140

Pro Pro Ala Ile Tyr Lys Ser Ala Cys Lys Ile Glu Val Lys Tyr Phe

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-continued

145	150	155	160
Pro Phe Asp Gln Gln Asn Cys Thr Leu Lys Phe Arg Ser Trp Thr Tyr	165	170	175
Asp His Thr Glu Ile Asp Met Val Leu Met Thr Pro Thr Ala Ser Met	180	185	190
Asp Asp Phe Thr Pro Ser Gly Glu Trp Asp Ile Val Ala Leu Pro Gly	195	200	205
Arg Arg Thr Val Asn Pro Gln Asp Pro Ser Tyr Val Asp Val Thr Tyr	210	215	220
Asp Phe Ile Ile Lys Arg Lys Pro Leu Phe Tyr Thr Ile Asn Leu Ile	225	230	235
Ile Pro Cys Val Leu Thr Thr Leu Leu Ala Ile Leu Val Phe Tyr Leu	245	250	255
Pro Ser Asp Cys Gly Glu Lys Met Thr Leu Cys Ile Ser Val Leu Leu	260	265	270
Ala Leu Thr Phe Phe Leu Leu Leu Ile Ser Lys Ile Val Pro Pro Thr	275	280	285
Ser Leu Asp Val Pro Leu Ile Gly Lys Tyr Leu Met Phe Thr Met Val	290	295	300
Leu Val Thr Phe Ser Ile Val Thr Ser Val Cys Val Leu Asn Val His	305	310	315
His Arg Ser Pro Ser Thr His Thr Met Ala Pro Trp Val Lys Arg Cys	325	330	335
Phe Leu His Lys Leu Pro Thr Phe Leu Phe Met Lys Arg Pro Gly Pro	340	345	350
Asp Ser Ser Pro Ala Arg Ala Phe Pro Pro Ser Lys Ser Cys Val Thr	355	360	365
Lys Pro Glu Ala Thr Ala Thr Ser Thr Ser Pro Ser Asn Phe Tyr Gly	370	375	380
Asn Ser Met Tyr Phe Val Asn Pro Ala Ser Ala Ala Ser Lys Ser Pro	385	390	395
Ala Gly Ser Thr Pro Val Ala Ile Pro Arg Asp Phe Trp Leu Arg Ser	405	410	415
Ser Gly Arg Phe Arg Gln Asp Val Gln Glu Ala Leu Glu Gly Val Ser	420	425	430
Phe Ile Ala Gln His Met Lys Asn Asp Asp Glu Asp Gln Ser Val Val	435	440	445
Glu Asp Trp Lys Tyr Val Ala Met Val Val Asp Arg Leu Phe Leu Trp	450	455	460
Val Phe Met Phe Val Cys Val Leu Gly Thr Val Gly Leu Phe Leu Pro	465	470	475
Pro Leu Phe Gln Thr His Ala Ala Ser Glu Gly Pro Tyr Ala Ala Gln	485	490	495
Arg Asp			

<210> SEQ ID NO 3
 <211> LENGTH: 705
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 3

Met Gln Pro Thr Ala Thr Met Ala Thr Ala Ala Thr Thr Thr Thr

-continued

1	5	10	15
Thr Thr Ala	Thr Val Ala Leu Thr Thr Ser Trp Asp Asn Ala Thr Gly		
	20	25	30
Arg Pro Thr	Ala Glu Pro Asp Pro Ile Leu Asp Asn Tyr Val Leu Leu		
	35	40	45
Val Val Val	Met Ser Leu Phe Val Gly Gly Thr Leu Val Val Leu Ser		
	50	55	60
Gly Val Leu	Leu Leu Cys Lys Arg Cys Trp Asp Val His Gln Arg Leu		
	65	70	75
Asn Arg Ala	Met Glu Glu Ala Glu Lys Thr Thr Thr Thr Tyr Leu Asp		
	85	90	95
Asn Gly Thr	His Pro Ala Gln Asp Pro Asp Phe Arg Gly Glu Asp Pro		
	100	105	110
Glu Cys Gln	Asp Ala Glu Thr Glu Arg Phe Leu Ser Thr Ser Ser Thr		
	115	120	125
Gly Arg Arg	Val Ser Phe Asn Glu Ala Ala Leu Phe Glu Gln Ser Arg		
	130	135	140
Lys Thr Gln	Asp Lys Gly Arg Arg Tyr Thr Leu Thr Glu Gly Asp Phe		
	145	150	155
His His Leu	Lys Asn Ala Arg Leu Thr His Leu His Leu Pro Pro Leu		
	165	170	175
Lys Ile Val	Thr Ile His Glu Cys Asp Ser Gly Glu Ala Ser Ser Ala		
	180	185	190
Thr Thr Pro	His Pro Ala Thr Ser Pro Lys Ala Thr Leu Ala Ile Phe		
	195	200	205
Gln Pro Pro	Gly Lys Ala Leu Thr Gly Arg Ser Val Gly Pro Ser Ser		
	210	215	220
Ala Leu Pro	Gly Asp Pro Tyr Asn Ser Ala Ala Gly Ala Thr Asp Phe		
	225	230	235
Ala Glu Ile	Ser Pro Ser Ala Ser Ser Asp Ser Gly Glu Gly Thr Ser		
	245	250	255
Leu Asp Ala	Gly Thr Arg Ser Thr Lys Ala Gly Gly Pro Gly Ala Ala		
	260	265	270
Ala Gly Pro	Gly Glu Ala Gly Pro Gly Ser Gly Ala Gly Thr Val Leu		
	275	280	285
Gln Phe Leu	Thr Arg Leu Arg Arg His Ala Ser Leu Asp Gly Ala Ser		
	290	295	300
Pro Tyr Phe	Lys Val Lys Lys Trp Lys Leu Glu Pro Ser Gln Arg Ala		
	305	310	315
Ala Ser Leu	Asp Thr Arg Gly Ser Pro Lys Arg His His Phe Gln Arg		
	325	330	335
Gln Arg Ala	Ala Ser Glu Ser Thr Glu Gln Glu Glu Gly Asp Ala Pro		
	340	345	350
Gln Glu Asp	Phe Ile Gln Tyr Ile Ala Arg Ala Gly Asp Ala Val Ala		
	355	360	365
Phe Pro His	Pro Arg Pro Phe Leu Ala Ser Pro Pro Pro Ala Leu Gly		
	370	375	380
Arg Leu Glu	Ala Ala Glu Ala Ala Gly Gly Ala Ser Pro Asp Ser Pro		
	385	390	395
Pro Glu Arg	Gly Ala Gly Ser Ala Gly Pro Glu Gln Gln Gln Pro Pro		
	405	410	415

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Leu Glu Pro Asp Ala Glu Arg Asp Ala Gly Pro Glu Gln Ala Gln Thr
 420 425 430
 Ser Tyr Arg Asp Leu Trp Ser Leu Arg Ala Ser Leu Glu Leu His Ala
 435 440 445
 Ala Ala Ser Asp His Ser Ser Ser Gly Asn Asp Arg Asp Ser Val Arg
 450 455 460
 Ser Gly Asp Ser Ser Gly Ser Gly Ser Gly Gly Ala Ala Pro Ala Phe
 465 470 475 480
 Pro Pro Pro Ser Pro Pro Ala Pro Arg Pro Lys Asp Gly Glu Ala Arg
 485 490 495
 Arg Leu Leu Gln Met Asp Ser Gly Tyr Ala Ser Ile Glu Gly Arg Gly
 500 505 510
 Ala Gly Asp Asp Thr Glu Pro Pro Ala Ala Pro Ala Arg Pro Arg Ser
 515 520 525
 Pro Arg Ala Trp Pro Arg Arg Pro Arg Arg Asp Tyr Ser Ile Asp Glu
 530 535 540
 Lys Thr Asp Ala Leu Phe His Glu Phe Leu Arg His Asp Pro His Phe
 545 550 555 560
 Asp Asp Thr Pro Ala Ala Ala Arg His Arg Ala Arg Ala His Pro His
 565 570 575
 Ala Arg Lys Gln Trp Gln Arg Gly Arg Gln His Ser Asp Pro Gly Ala
 580 585 590
 Arg Ala Ala Pro Ala Leu Ala Gly Thr Pro Ala Pro Pro Ala Gly Ala
 595 600 605
 Ala Arg Pro Ala Arg Ala Pro Leu Arg Arg Gly Asp Ser Val Asp Gly
 610 615 620
 Pro Pro Asp Gly Arg Thr Leu Gly Gly Ala Gly Asp Asp Pro Ala Ile
 625 630 635 640
 Pro Val Ile Glu Glu Glu Pro Gly Gly Gly Gly Cys Pro Gly Ser Gly
 645 650 655
 Leu Cys Val Leu Pro Ser Gly Ser Val Leu Asp Lys Leu Ala Ala Gly
 660 665 670
 Leu Asp Glu Arg Leu Phe Pro Pro Arg Leu Ala Glu Pro Val Val Ala
 675 680 685
 Thr Pro Ala Leu Val Ala Ala Ala Pro Thr Ser Pro Asp His Ser Pro
 690 695 700
 Ala
 705

<210> SEQ ID NO 4
 <211> LENGTH: 630
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 4

Met Ala Ala Lys Thr Pro Ser Ser Glu Glu Ser Gly Leu Pro Lys Leu
 1 5 10 15
 Pro Val Pro Pro Leu Gln Gln Thr Leu Ala Thr Tyr Leu Gln Cys Met
 20 25 30
 Arg His Leu Val Ser Glu Glu Gln Phe Arg Lys Ser Gln Ala Ile Val
 35 40 45
 Gln Gln Phe Gly Ala Pro Gly Gly Leu Gly Glu Thr Leu Gln Gln Lys

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50					55					60					
Leu 65	Leu	Glu	Arg	Gln	Glu 70	Lys	Thr	Ala	Asn	Trp 75	Val	Ser	Glu	Tyr	Trp 80
Leu	Asn	Asp	Met	Tyr 85	Leu	Asn	Asn	Arg	Leu 90	Ala	Leu	Pro	Val	Asn 95	Ser
Ser	Pro	Ala	Val	Ile	Phe	Ala	Arg	Gln 105	His	Phe	Pro	Gly	Thr 110	Asp	Asp
Gln	Leu	Arg	Phe	Ala	Ala	Ser	Leu 120	Ile	Ser	Gly	Val	Leu	Ser	Tyr	Lys
Ala	Leu 130	Leu	Asp	Ser	His	Ser 135	Ile	Pro	Thr	Asp	Cys 140	Ala	Lys	Gly	Gln
Leu 145	Ser	Gly	Gln	Pro	Leu 150	Cys	Met	Lys	Gln	Tyr 155	Tyr	Gly	Leu	Phe	Ser 160
Ser	Tyr	Arg	Leu	Pro 165	Gly	His	Thr	Gln	Asp 170	Thr	Leu	Val	Ala	Gln 175	Asn
Ser	Ser	Ile	Met	Pro	Glu	Pro	Glu 185	His	Val	Ile	Val	Ala	Cys 190	Cys	Asn
Gln	Phe 195	Phe	Val	Leu	Asp	Val	Val 200	Ile	Asn	Phe	Arg	Arg 205	Leu	Ser	Glu
Gly 210	Asp	Leu	Phe	Thr	Gln	Leu 215	Arg	Lys	Ile	Val	Lys 220	Met	Ala	Ser	Asn
Glu 225	Asp	Glu	Arg	Leu	Pro 230	Pro	Ile	Gly	Leu	Leu 235	Thr	Ser	Asp	Gly	Arg 240
Ser	Glu	Trp	Ala	Glu 245	Ala	Arg	Thr	Val	Leu 250	Val	Lys	Asp	Ser	Thr 255	Asn
Arg	Asp	Ser	Leu	Asp 260	Met	Ile	Glu	Arg 265	Cys	Ile	Cys	Leu	Val 270	Cys	Leu
Asp	Ala 275	Pro	Gly	Gly	Val	Glu	Leu 280	Ser	Asp	Thr	His	Arg 285	Ala	Leu	Gln
Leu 290	Leu	His	Gly	Gly	Gly	Tyr 295	Ser	Lys	Asn	Gly	Ala 300	Asn	Arg	Trp	Tyr
Asp 305	Lys	Ser	Leu	Gln	Phe 310	Val	Val	Gly	Arg	Asp 315	Gly	Thr	Cys	Gly	Val 320
Val	Cys	Glu	His	Ser 325	Pro	Phe	Asp	Gly	Ile 330	Val	Leu	Val	Gln	Cys 335	Thr
Glu	His 340	Leu	Leu	Lys	His	Met	Thr	Gln 345	Ser	Ser	Arg	Lys 350	Leu	Ile	Arg
Ala	Asp 355	Ser	Val	Ser	Glu	Leu	Pro 360	Ala	Pro	Arg	Arg	Leu 365	Arg	Trp	Lys
Cys 370	Ser	Pro	Glu	Ile	Gln	Gly 375	His	Leu	Ala	Ser	Ser 380	Ala	Glu	Lys	Leu
Gln 385	Arg	Ile	Val	Lys	Asn	Leu 390	Asp	Phe	Ile	Val 395	Tyr	Lys	Phe	Asp	Asn 400
Tyr	Gly	Lys	Thr	Phe 405	Ile	Lys	Lys	Gln	Lys 410	Cys	Ser	Pro	Asp	Ala 415	Phe
Ile	Gln	Val	Ala	Leu 420	Gln	Leu	Ala	Phe	Tyr 425	Arg	Leu	His	Arg 430	Arg	Leu
Val	Pro	Thr	Tyr	Glu 435	Ser	Ala	Ser 440	Ile	Arg	Arg	Phe	Gln 445	Glu	Gly	Arg
Val 450	Asp	Asn	Ile	Arg	Ser	Ala 455	Thr	Pro	Glu	Ala 460	Leu	Ala	Phe	Val	Arg

-continued

Ala Val Thr Asp His Lys Ala Ala Val Pro Ala Ser Glu Lys Leu Leu
 465 470 475 480

Leu Leu Lys Asp Ala Ile Arg Ala Gln Thr Ala Tyr Thr Val Met Ala
 485 490 495

Ile Thr Gly Met Ala Ile Asp Asn His Leu Leu Ala Leu Arg Glu Leu
 500 505 510

Ala Arg Ala Met Cys Lys Glu Leu Pro Glu Met Phe Met Asp Glu Thr
 515 520 525

Tyr Leu Met Ser Asn Arg Phe Val Leu Ser Thr Ser Gln Val Pro Thr
 530 535 540

Thr Thr Glu Met Phe Cys Cys Tyr Gly Pro Val Val Pro Asn Gly Tyr
 545 550 555 560

Gly Ala Cys Tyr Asn Pro Gln Pro Glu Thr Ile Leu Phe Cys Ile Ser
 565 570 575

Ser Phe His Ser Cys Lys Glu Thr Ser Ser Ser Lys Phe Ala Lys Ala
 580 585 590

Val Glu Glu Ser Leu Ile Asp Met Arg Asp Leu Cys Ser Leu Leu Pro
 595 600 605

Pro Thr Glu Ser Lys Pro Leu Ala Thr Lys Glu Lys Ala Thr Arg Pro
 610 615 620

Ser Gln Gly His Gln Pro
 625 630

<210> SEQ ID NO 5

<211> LENGTH: 365

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 5

Met Asp Gly Pro Ala Glu Pro Gln Ile Pro Gly Leu Trp Asp Thr Tyr
 1 5 10 15

Glu Asp Asp Ile Ser Glu Ile Ser Gln Lys Leu Pro Gly Glu Tyr Phe
 20 25 30

Arg Tyr Lys Gly Val Pro Phe Pro Val Gly Leu Tyr Ser Leu Glu Ser
 35 40 45

Ile Ser Leu Ala Glu Asn Thr Gln Asp Val Arg Asp Asp Asp Ile Phe
 50 55 60

Ile Ile Thr Tyr Pro Lys Ser Gly Thr Thr Trp Met Ile Glu Ile Ile
 65 70 75 80

Cys Leu Ile Leu Lys Glu Gly Asp Pro Ser Trp Ile Arg Ser Val Pro
 85 90 95

Ile Trp Glu Arg Ala Pro Trp Cys Glu Thr Ile Val Gly Ala Phe Ser
 100 105 110

Leu Pro Asp Gln Tyr Ser Pro Arg Leu Met Ser Ser His Leu Pro Ile
 115 120 125

Gln Ile Phe Thr Lys Ala Phe Phe Ser Ser Lys Ala Lys Val Ile Tyr
 130 135 140

Met Gly Arg Asn Pro Arg Asp Val Val Val Ser Leu Tyr His Tyr Ser
 145 150 155 160

Lys Ile Ala Gly Gln Leu Lys Asp Pro Gly Thr Pro Asp Gln Phe Leu
 165 170 175

Arg Asp Phe Leu Lys Gly Glu Val Gln Phe Gly Ser Trp Phe Asp His

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180							185					190				
Ile	Lys	Gly	Trp	Leu	Arg	Met	Lys	Gly	Lys	Asp	Asn	Phe	Leu	Phe	Ile	
	195						200					205				
Thr	Tyr	Glu	Glu	Leu	Gln	Gln	Asp	Leu	Gln	Gly	Ser	Val	Glu	Arg	Ile	
	210					215					220					
Cys	Gly	Phe	Leu	Gly	Arg	Pro	Leu	Gly	Lys	Glu	Ala	Leu	Gly	Ser	Val	
225					230					235				240		
Val	Ala	His	Ser	Thr	Phe	Ser	Ala	Met	Lys	Ala	Asn	Thr	Met	Ser	Asn	
				245					250					255		
Tyr	Thr	Leu	Leu	Pro	Pro	Ser	Leu	Leu	Asp	His	Arg	Arg	Gly	Ala	Phe	
		260						265					270			
Leu	Arg	Lys	Gly	Val	Cys	Gly	Asp	Trp	Lys	Asn	His	Phe	Thr	Val	Ala	
	275						280					285				
Gln	Ser	Glu	Ala	Phe	Asp	Arg	Ala	Tyr	Arg	Lys	Gln	Met	Arg	Gly	Met	
	290					295					300					
Pro	Thr	Phe	Pro	Trp	Asp	Glu	Asp	Pro	Glu	Glu	Asp	Gly	Ser	Pro	Asp	
305					310					315				320		
Pro	Glu	Pro	Ser	Pro	Glu	Pro	Glu	Pro	Lys	Pro	Ser	Leu	Glu	Pro	Asn	
				325					330					335		
Thr	Ser	Leu	Glu	Arg	Glu	Pro	Arg	Pro	Asn	Ser	Ser	Pro	Ser	Pro	Ser	
		340					345					350				
Pro	Gly	Gln	Ala	Ser	Glu	Thr	Pro	His	Pro	Arg	Pro	Ser				
	355						360					365				

What is claimed is:

1. An isolated recombinant cell comprising:
 - a) a heterologous nucleic acid encoding an $\alpha 6$ subunit of an $\alpha 6\beta 4$ nicotinic acetylcholine receptor (nAChR);
 - b) a heterologous nucleic acid encoding a $\beta 4$ subunit of an $\alpha 6\beta 4$ nAChR; and
 - c) at least one heterologous nucleic acid selected from the group consisting of a heterologous nucleic acid encoding beta associated regulatory protein (BARP), a heterologous nucleic acid encoding choline O-acetyltransferase (CHAT), and a heterologous nucleic acid encoding sulfotransferase family 2B member 1 (SULT2B1).
2. The isolated recombinant cell of claim 1, wherein the recombinant cell comprises the heterologous nucleic acid encoding BARP, the heterologous nucleic acid sequence encoding CHAT, and the heterologous nucleic acid sequence encoding SULT2B1.
3. The isolated recombinant cell of claim 1, wherein the recombinant cell is a mammalian cell.
4. The isolated recombinant cell of claim 3, wherein the mammalian cell is selected from the group consisting of a human embryonic kidney 293T (HEK293T) cell, a HEK293F cell, a HeLa cell, a Chinese hamster ovary (CHO) cell, a NIH 3T3 cell, a MCF-7 cell, a Hep G2 cell, a baby hamster kidney (BHK) cell, and a Cos7 cell.
5. The isolated recombinant cell of claim 1, wherein the $\alpha 6$ subunit of the $\alpha 6\beta 4$ nAChR comprises an amino acid sequence with at least 95% identity to the amino acid sequence of SEQ ID NO: 1.
6. The isolated recombinant cell of claim 1, wherein the $\beta 4$ subunit of the $\alpha 6\beta 4$ nAChR comprises an amino acid sequence with at least 95% identity to the amino acid sequence of SEQ ID NO: 2.
7. The isolated recombinant cell of claim 1, wherein the BARP comprises an amino acid sequence with at least 95% identity to the amino acid sequence of SEQ ID NO: 3.
8. The isolated recombinant cell of claim 1, wherein the CHAT comprises an amino acid sequence with at least 95% identity to the amino acid sequence of SEQ ID NO: 4.
9. The isolated recombinant cell of claim 1, wherein the SULT2B1 comprises an amino acid sequence with at least 95% identity to the amino acid sequence of SEQ ID NO: 5.
10. A method for identifying agonists, antagonists, or positive allosteric modulators of $\alpha 6\beta 4$ nAChR, the method comprising:
 - a) contacting the isolated recombinant cell of claim 1 with an agent; and
 - b) determining the activity of the $\alpha 6\beta 4$ nAChR of the isolated recombinant cell, wherein the agent is identified as an agonist or positive allosteric modulator (PAM) if the agent enhances the activity of the $\alpha 6\beta 4$ nAChR and the agent is identified as an antagonist if the agent decreases the activity of the $\alpha 6\beta 4$ nAChR as compared to the activity of the $\alpha 6\beta 4$ nAChR when the isolated recombinant cell was not contacted with the agent.
11. The method of claim 10, wherein step b) comprises determining calcium flux of the isolated recombinant cell, wherein the agent is identified as an agonist if the agent enhances the calcium flux as compared to the calcium flux when the isolated recombinant cell was not contacted with the agent.

12. The method of claim **10**, wherein step b) comprises determining calcium flux and nicotine-evoked calcium flux of the isolated recombinant cell, wherein the agent is identified as an PAM if the agent does not enhance calcium flux and enhances the nicotine-evoked calcium flux as compared to the calcium flux and nicotine-evoked calcium flux when the isolated recombinant cell was not contacted with the agent

13. The method of claim **10**, wherein step b) comprises determining nicotine-evoked calcium flux of the isolated recombinant cell, wherein the agent is identified as an antagonist if the agent decreases the nicotine-evoked calcium flux as compared to the nicotine-evoked calcium flux when the isolated recombinant cell was not contacted with the agent.

14. The method of claim **10**, wherein the isolated recombinant cell is incubated at about 25° C.-35° C. for about 20-50 hours prior to being contacted with the agent.

15. The method of claim **10**, wherein the agent is a small molecule or peptide.

16. A kit comprising (i) the isolated recombinant cell of claim **1**, and (ii) instructions for use.

* * * * *